MedRule Prior-Authorization Decision Support

Interactive Clinical Evaluation & Reviewer Walkthrough Guide

Live Cloud URL: https://medrule-preauth.onrender.com	**Architecture:** 100% Deterministic (Zero Generative AI)
GitHub Repository: github.com/shivarajsg/PreAuth	**Knowledge Base:** 372 Policy Rules (9 Specialties)
Health Status: **OPERATIONAL & HEALTHY (200 OK)**	**Safety Invariant:** Exactly 0 Approved Rules (Strictly Enforced)

Cloud Container Notice: Hosted on a containerized cloud runtime. If idle, the initial page load may require 15–25 seconds while the instance spins up. All subsequent interactions, rule searches, and case evaluations will execute immediately.

STEP 1 System Dashboard & Governance Invariants

Direct URL: https://medrule-preauth.onrender.com/dashboard

- **Rule Knowledge Base Audit:** Inspect live breakdown of 372 rules across 9 clinical specialties (322 in Draft, 50 in Needs Human Review, 0 in Approved).
- **Zero-Approved Safety Invariant:** Enforces that no rule can be auto-approved or executed in binding mode without formal multi-clinician sign-off.
- **Persona Switcher (Top Right):** Switch personas to test Role-Based Access Control (RBAC):
  - **Dr. Sarah Chen, MD:** Clinical SME Reviewer (Imaging & Diagnostics).
  - **Dr. Eleanor Vance, MD:** Medical Director (Authorized Sign-Off & Approvals).
  - **Quality Auditor:** Read-Only Compliance & Invariant Monitoring.

STEP 2 Rule Explorer & Clinical Policy Workbench

Direct URL: https://medrule-preauth.onrender.com/rules

- **Domain Filtering:** Filter rules across Cardiology, Radiology, Oncology, Musculoskeletal, DME, and 4 other specialties.
- **CPT/HCPCS Search:** Search for specific codes or clinical conditions (e.g., CPT 70450 for CT Head or CPT 93306 for Echocardiography).
- **Interactive Workbench:** Click any rule (e.g., RULE-RAD-001) to inspect:
  - **Structured Logic:** Deterministic boolean logic trees (AND/OR condition groups and numerical thresholds).
  - **Source Provenance:** Exact verbatim quotation from governing payer policies with physical page citations.

STEP 3 Human Review Queue & Governance Workflow

Direct URL: https://medrule-preauth.onrender.com/queue

- **Triage Ambiguous Criteria:** Inspect the 50 rules requiring clinical human clarification due to conflicting guidelines or edge cases.
- **Propose Criteria Corrections:** Submit clinical correction diffs with documented rationale and supporting medical literature citations.
- **Enforced Approval Prerequisites:** Validates the 8 mandatory prerequisites (dual clinician sign-off, citation verification, audit hashing) before any promotion.

STEP 4 Case Intake & Live Deterministic Evaluation

Direct URL: https://medrule-preauth.onrender.com/intake

- **One-Click Sample Case:** Click **'Load Sample Case'** to populate a realistic synthetic prior-authorization request (or choose a clinical preset).
- **Execute Evaluation:** Click **'Run Intake & Evaluate'** to execute the 10-stage deterministic pipeline:
  - 1. Member eligibility & benefit verification (simulated connector).
  - 2. Prior-authorization requirement determination.
  - 3. Policy routing to the exact clinical guideline.
  - 4. Deterministic condition evaluation (MET, NOT_MET, or UNKNOWN).
  - 5. Comprehensive audit generation with full citation trace and non-binding CDS disclaimer.

STEP 5 Pilot Adjudication & Concordance Tracking

Direct URL: https://medrule-preauth.onrender.com/pilot

- **Synthetic Cohort Benchmark:** Compare automated decision-support outputs against benchmark human adjudication across 36 pilot cases.
- **Discrepancy Root-Cause Analysis:** Inspect discrepancy categories: documentation ambiguity, threshold mismatch, or missing clinical notes.

STEP 6 System Health & Safety Invariant Telemetry

Direct URL: https://medrule-preauth.onrender.com/api/health

- **Live JSON Health Status:** Direct machine-readable verification endpoint reporting:
  - **database:** CONNECTED (SQLite with cryptographic hash-chain audit logging).
  - **zero_approved_invariant:** true (strict enforcement active).
  - **is_binding:** false (all evaluations remain strictly non-binding CDS).
  - **submission_adapter:** DISABLED (external payer networks permanently locked).

Reviewer Verification Checklist

Verification Item	Expected Behavior & Invariant	Status
Zero Generative AI Invariant	100% deterministic boolean logic trees; zero hallucinated thresholds or probabilistic weights.	**ENFORCED**
Non-Binding Evaluation	Every intake evaluation asserts is_binding: false with complete audit trail.	**ENFORCED**
Zero Approved Rules	Exactly 0/372 rules in APPROVED status (322 Draft, 50 Needs Human Review).	**ENFORCED**
Outbound Submission Locking	External submission adapter locked (DISABLED); eligibility & benefits run in mock mode.	**LOCKED**